

VLAN Concepts Theory: VLANs: Segmentation, benefits, and use cases. Practical : Create VLANs on a switch and assign ports. Task: Document VLAN configurations.

Q-1 On a Cisco Packet Tracer switch, create two VLANs named 'Engineering' (VLAN 10) and 'Marketing' (VLAN 20) using the CLI commands.

[A]

- To understand VLAN segmentation, create VLANs on a Cisco switch, assign switch ports to different VLANs, document the configuration, and understand the benefits of VLANs in a college campus network.

○ **Create Two VLANs :**

VLAN ID	VLAN Name
10	Engineering
20	Marketing

➤ **Cisco Packet Tracer CLI Commands**

Enter privileged mode:

Switch> enable

Switch# configure terminal

➤ **Create VLAN 10 — Engineering**

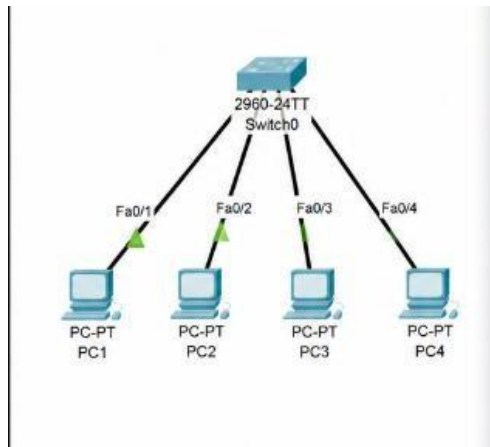
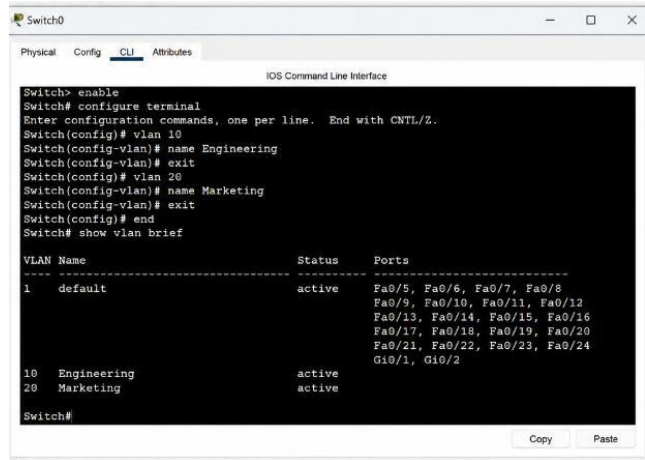
Switch(config)# vlan 10

Switch(config-vlan)# name Engineering

Switch(config-vlan)# exit

➤ **Create VLAN 20 — Marketing**

```
Switch(config)# vlan 20
Switch(config-vlan)# name Marketing
Switch(config-vlan)# exit
```

```
Switch0
Physical Config CLI Attributes
IOS Command Line Interface

Switch> enable
Switch# configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)# vlan 10
Switch(config-vlan)# name Engineering
Switch(config-vlan)# exit
Switch(config)# vlan 20
Switch(config-vlan)# name Marketing
Switch(config-vlan)# exit
Switch(config)# end
Switch# show vlan brief

VLAN Name                Status    Ports
-----
1    default                active    Fa0/5, Fa0/6, Fa0/7, Fa0/8
          Fa0/9, Fa0/10, Fa0/11, Fa0/12
          Fa0/13, Fa0/14, Fa0/15, Fa0/16
          Fa0/17, Fa0/18, Fa0/19, Fa0/20
          Fa0/21, Fa0/22, Fa0/23, Fa0/24
          Gi0/1, Gi0/2
10   Engineering            active
20   Marketing              active

Switch#
```

Q-2 Assign ports FastEthernet0/1 and FastEthernet0/2 to VLAN 10, and ports FastEthernet0/3 and FastEthernet0/4 to VLAN 20 on your switch.

[A]

➤ **Assign Fa0/1 and Fa0/2 to VLAN 10**

```
Switch(config)# interface range fastEthernet 0/1 - 2
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 10
Switch(config-if-range)# exit
```

➤ **Assign Fa0/3 and Fa0/4 to VLAN 20**

```
Switch(config)# interface range fastEthernet 0/3 - 4
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 20
Switch(config-if-range)# exit
Save the configuration: Switch(config)# end
Switch# write memory or:
```

Switch# copy running-config startup-config

Q-3 Write a short document (in a text file) listing the VLAN IDs, names, and the ports assigned to each VLAN from your configuration.

[A]

➤ You can put the following content into a text file such as **VLAN_Configuration.txt**:

VLAN CONFIGURATION DOCUMENT

=====

Switch: Cisco Packet Tracer Switch

VLAN 10

VLAN Name : Engineering

Ports : FastEthernet0/1, FastEthernet0/2

VLAN 20

VLAN Name : Marketing

Ports : FastEthernet0/3, FastEthernet0/4

Summary

VLAN 10 - Engineering - Fa0/1, Fa0/2

VLAN 20 - Marketing - Fa0/3, Fa0/4

➤ **Configuration Summary**

VLAN ID	VLAN Name	Assigned Ports
10	Engineering	Fa0/1, Fa0/2
20	Marketing	Fa0/3, Fa0/4

Q-4 Explain, in your own words, two benefits of using VLANs in a college campus network that has separate labs for coding and networking.

[A]

➤ **Benefit 1: Network Segmentation**

- VLANs can separate different departments or labs into different logical networks.

For example:

Coding Lab → VLAN 10 Networking Lab → VLAN 20

- Devices in the Coding Lab remain logically separated from devices in the Networking Lab. This reduces unnecessary broadcast traffic and makes the network easier to manage.

➤ **Benefit 2: Improved Security**

- VLANs can help restrict communication between different groups.
For example, computers in the Coding Lab do not automatically need direct access to all computers in the Networking Lab. Communication between VLANs can be controlled using a **router or Layer 3 switch** and appropriate access-control rules.
- This helps protect important systems and reduces the possibility of unauthorized access.

